Full-State Feedback Controller Design for Pendulum System

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***Abstract*— In this lab, a full-state feedback controller was designed for a rotational pendulum system, using both manual tuning and Linear Quadratic Regulator (LQR) methods. The platen and pendulum error and their derivatives represent the states of the controller which form the full-state vector, which the controller uses to compute the feedback control input. The LQR's Q matrix and the R value were modified to adjust the controller's effort. The performances of control grains were quantified by the Root Mean Squared of Tracking Error (RMSTE) of the platen and pendulum rotational position, Root Mean Squared (RMS) of Command values, rise time, and maximum pendulum angle. Using these metrics, it was found that the LQR method had a 13.8% decrease in RMSTE platen rotational position compared to the manual tuning method, making it the more effective method. The smallest RMSTE of platen rotational position came from the LQR controller with Q values of 70, 40, 0.01, & 0.01 in order of the system's states with an R value of 1. Additionally, modifying R to be 0.1 increased the RMS command to 1126 bits while an R value of 10 decreased the RMS command to 106.2 bits.**

***Index Terms*— Full-state Feedback Controller, Linear Quadratic Regulation (LQR), Manual Tuning, Rotational Pendulum**

I. Introduction

THE objective of this lab is the design of a full-state feedback controller for a rotational pendulum system and the implementation of various controllers under multiple test conditions. Manual gain tuning and Linear Quadratic Regulation (LQR) methods were used to gauge the system's performance, and the effects of modifying the Q matrix and the R value were also examined [1].

The Furuta pendulum used in this lab is made up of a pendulum attached to a motor-powered arm, which rotates in the vertical and horizontal planes respectively, displayed in (Fig. 1). The system captures linear and nonlinear control behaviors, and the full dynamic model of a Furuta pendulum was derived by Cazzolato and Prime, who used both a Lagrangian formulation and a Newton-Euler approach. They provided equations to approximate the full dynamics of the system, including a linearized state-space representation [2]. The system parameters of motor resistance, R_m, and motor torque constant, K_m, were calculated from values of motor voltage, stall torque, and stall current. The mass of the pendulum, $m_{pendulum}$, was measured using a scale, and the lengths of both the platen, L_1, and the pendulum, L_2, were found using a tape measure. A velocity experiment using a square

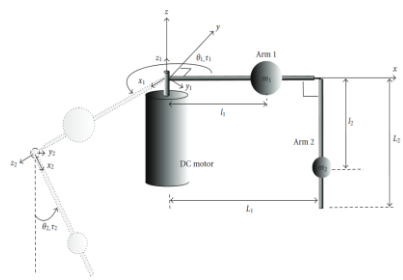

Fig. 1. Schematic of Cazzolato and Prime single rotary inverted pendulum system

wave at frequency of 0.25 Hz was used to determine the base friction coefficient, b_1, and the system inertia, motor torque constant, $\hat{J}_o$. A free-swing experiment was used to find the pendulum arm friction coefficient, b_2. This experiment involved releasing the pendulum at a 20-degree angle and recording the oscillations until the pendulum came to rest. These values were used to determine the A (1), B (2), C (3) and D (4) matrices based on Cazzolato and Prime's dynamic models.

TABLE I
SYSTEM PARAMETERS OF PENDULUM SYSTEM

Symbol	Value	Units
R_m	16.0	Ω
K_m	0.526	N·m/A
$m_{pendulum}$	0.008	kg
L_1	0.100	m
L_2	0.149	m
l_2	0.102	m
b_1	0.015	N·m·s/rad
b_2	1.88 E-4	kg·m²/s
$\hat{J}_0$	0.005	kg · m²
$\hat{J}_2$	1.00 E-4	kg · m²

These parameters were taken from [3].

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1.11 & -6.01 & 0.026 \\ 0 & -63.6 & 4.56 & -1.52 \end{bmatrix} \quad (1)$$

$$B = \begin{bmatrix} 0 \\ 0 \\ 6.15 \\ -4.65 \end{bmatrix} \tag{2}$$

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \tag{3}$$

$$D = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \tag{4}$$

For this lab, LQR methods were used to design a full-state feedback controller. The goal was to create a K gain matrix that minimizes the cost function, J, consisting of the Q matrix and the R value, along with x and u, which are a part of the state space equations, shown in (5).

$$J = \int_{t_o}^{t_f} (x^T Q x + u^T R u) dt \tag{5}$$

The Q matrix shown in (6) is defined by the inverse of the error of the platen rotation angle, θ_1, the pendulum rotation angle, θ_2, the platen rotation derivative of the platen rotational angle, $\dot{\theta}_1$, and the pendulum rotation derivative of the pendulum rotational angle, $\dot{\theta}_2$. The values in the Q matrix are inversely proportional to the state error. R is a matrix that controls the system's effort in achieving the desired output values of the system. Values for the Q matrix and R were selected to optimize the system and find the optimal control gains.

$$Q = \begin{bmatrix} 1/e_{\theta_1} & 0 & 0 & 0 \\ 0 & 1/e_{\theta_2} & 0 & 0 \\ 0 & 0 & 1/e_{\dot{\theta}_1} & 0 \\ 0 & 0 & 0 & 1/e_{\dot{\theta}_2} \end{bmatrix} \tag{6}$$

Equation (7) is the full-state feedback law where u is the input to the system based on the K gain values and the system's state values and is used in the LQR cost function to control the Furuta pendulum system's performance.

$$u = -Kx \tag{7}$$

II. Procedure

Hardware Setup

A Furuta pendulum setup was used for this experiment. It consisted of a driven arm that rotates in the horizontal plane and a pendulum arm attached to the driven arm that rotates in the vertical plane. The pendulum was connected to the pendulum encoder that was fixed to the driven arm, via a breakaway pendulum socket. The drive arm was part of the platen assembly and was equipped with a driven gear, where it was meshed to a drive gear rotated by a gear motor. The motor was controlled by the pendulum driver. The Digital Acquisition Device (DAQ) was connected to the pendulum driver via EXT connection [4]. The platen and pendulum rotation encoders fastened to the top of the assembly generated electric pulses as the shaft was turned, where the pulse count correlated to angular position [5].

LabVIEW VI Setup

LabVIEW was the graphical programming environment software used to collect data from the experiments. The Virtual Instrument (VI) setup from the previous lab was modified to fit this experiment's data collection requirements [3]. Inputs for 4 gains, K_1, K_2, K_3, and K_4 corresponding to the 4 states of the full-state feedback controller, θ_1, θ_2, $\dot{\theta}_1$, and $\dot{\theta}_2$ were added to the VI to control the system performance of the Furuta pendulum. A square wave was used for all experiments with the desired rotational amplitude set at ± 60 degrees oscillating at a frequency of 0.25 Hz. The turn on command was set at 0 bits. Desired platen rotation angle, actual platen rotation angle, platen rotational velocity, pendulum angle, elapsed time in seconds, command value, and platen and pendulum positional errors were recorded. After collecting, the data was processed into Root Mean Squared of Tracking Error (RMSTE) from tracking error values of the platen and pendulum, and Root Mean Squared (RMS) from the Command values to compare each manual tuning and gain cases of LQR performance. All tasks were done as a pendulum-down tracking system. Equation (8) is the RMS equation where, x_n, is a data value and N is the number of data values.

$$X_{RMS} = \sqrt{\frac{(x_1^2 + x_2^2 \dots x_n^2)}{N}} \tag{8}$$

Feedback Gains

Manual tuning was first used to determine the K_1 and K_3 values by tracking a square wave and tuning the gains to make the platen rotations as accurately as possible to the desired platen rotation value. The pendulum during this task was immobilized using tape to focus on the platen rotation system performance, and K_2 and K_4 were set to 0 and the system performance was recorded. After K_1 and K_3 were determined, the pendulum was free to rotate, and the system performance was recorded again with the determined gain values.

To analyze the system performance with the pendulum input, K_1 and K_3 were first set to 0 and K_2 and K_4 were tuned manually to minimize the pendulum motion while ensuring the platen assembly followed the rotational amplitude as accurately as possible. Then K_1 and K_3 were also changed to achieve the target performance metrics.

The LQR method was used to design a full-state feedback controller for the single motor input, multiple rotational output system. An R value of 1 was used to focus on the system performance controlled by the Q matrix. For the first LQR case, the Q matrix shown in (8) gave more weight to platen rotation accuracy and less weight to the pendulum motion during the rotation. The velocity states were given minimal weights to focus on the platen and pendulum position states' performance. Using MATLAB's lqr() function, the gains were calculated using the A, B, Q, and R matrices.

$$Q = \begin{bmatrix} 70 & 0 & 0 & 0 \\ 0 & 40 & 0 & 0 \\ 0 & 0 & 0.01 & 0 \\ 0 & 0 & 0 & 0.01 \end{bmatrix} \quad (9)$$

In the second LQR case, the Q values were changed to determine a system performance that minimizes pendulum angle deviations during base-link tracking and performance was recorded. To show the R value's actuator control, Case 3 had an R value of 0.1 and Case 4 had an R value of 10, where the resulting system performance was tested and recorded.

III. Results

Test Condition Results

Table II presents the feedback gains of each test condition, including manual tuning and 4 modified LQR cases. Case 1, 3, and 4 used Q values of 70 and 40, for platen and pendulum rotational positional states respectively. R values of 1, 0.1, and 10 were used by each Case respectively. Case 2 used Q values of 30 and 200 with an R value of 1. Fig. 3 through 8 display the systems' response for each condition. The LQR method was most effective in reducing RMTSE platen values with Case 1 and Case 2 having the lowest values. The fastest rise time was Case 3 with a 0.064 second rise time with the highest maximum pendulum angle. The manual tuning of the K gains had the best control for the RMSTE pendulum angles with a value of 8.40 degrees. Using the impulse function in MATLAB, matrices A, B, C and D were used to plot the impulse response shown in (Fig. 2). The step response stabilizes without overshooting and the oscillations dampen to zero which was the expected behavior.

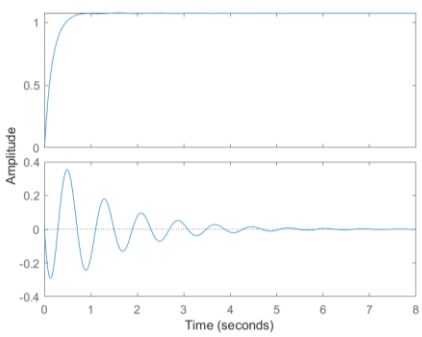

Fig. 2. Impulse response plot by MATLAB from A, B, C, and D matrix values.

TABLE II
Feedback Gain, RMSTE, RMS Command, and Performance Metrics

Method/ Gains	RMSTE Platen (°)	RMSTE Pendulum (°)	RMS Command (bits)	Rise Time (s)	Maximum Pendulum Angle (°)
Untaped: K_1 = -14.1 K_2 = -10 K_3 = 0 K_4 = 0.02	35.5	40.6	350	0.408	69.9
Manual Tune: K_1 = -4.81 K_2 = -10 K_3 = -0.01 K_4 = -0.01	38.4	8.40	160.7	0.496	26.7

TABLE II Continued

Method/ Gains	RMSTE Platen (°)	RMSTE Pendulum (°)	RMS Command (bits)	Rise Time (s)	Maximum Pendulum Angle (°)
LQR Case 1: K_1 = -12.4486 K_2 = -4.6143 K_3 =1.6458 K_4 = -0.0051	33.1	23.3	230	0.483	50.1
LQR Case 2: K_1 = -8.1495 K_2 = -17.2448 K_3 = 1.6683 K_4 = -0.6047	37.7	13.29	202	0.497	40.3
LQR Case 3: K_1 = -39.366 K_2 = -15.3406 K_3 = 4.5733 K_4 = 1.1672	87.1	34.2	1126	0.064	77.4
LQR Case 4: K_1 = -3.9366 K_2 = -0.6772 K_3 = 0.5719 K_4 = -0.0368	47.3	2.94	106.2	0.388	26.7

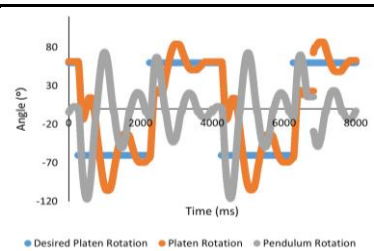

Fig. 3. Untaped pendulum system, pendulum resulted in erratic movements

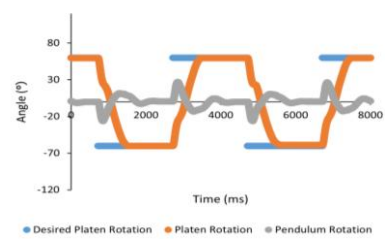

Fig. 4. Manually tuned system, pendulum movement is more stable

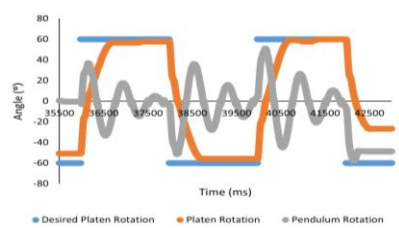

Fig. 5. LQR tuned system: Case 1, platen rotation accuracy was weighed more than pendulum stability

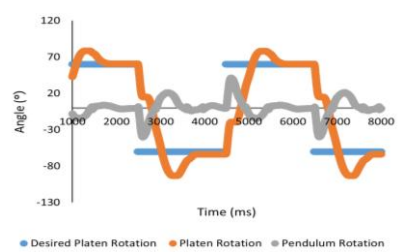

Fig. 6. LQR tuned system: Case 2, pendulum rotation stability was weighted more than platen rotation accuracy

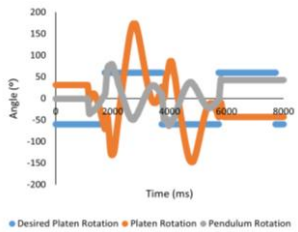

Fig. 7. LQR tuned system: Case 3, R matrix value of 0.1 system performance test. The system is unstable and does not match the desired rotation.

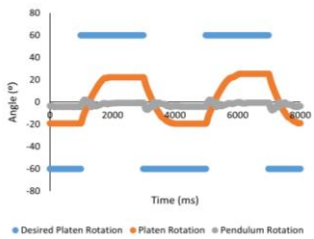

Fig. 8. LQR tuned system: Case 4, R matrix value of 10 system performance test. Both the platen and the pendulum undershoot the desired rotation.

IV. Discussion

The goal of this lab was to design a full-state feedback controller for a pendulum system using both manual tuning and LQR methods. The LQR method was most effective in reducing RMSTE platen values with Case 1 and Case 2 having the lowest values with the chosen Q matrix values. The fastest rise time was Case 3 with a 0.064 second rise time with the highest pendulum angle. The manual tuning of the K gains had the best control for the RMSTE pendulum angles with 8.40 degrees.

Manual Gain Tuning vs LQR

The manual tuning method led to a less effective platen rotation and a less effective pendulum rotation (Fig. 4) relative to the LQR Case 1 method (Fig. 5). The RMSTE platen was 38.4° for the manual tuning method, while being 33.1° for the LQR method. This represents a 13.8% decrease. The RMSTE pendulum was 8.40° for the manual tuning method, and 23.3° for the LQR method, representing a 177.4% increase. These results were expected since the LQR optimizes the state-feedback gains by minimizing the cost function which allows for tighter error control compared to the trial-and-error method of manual tuning. Specifically, the first index of the Q value was set at 70, restricting the error to 0.7°. The RMSTE pendulum increased due to the K_2 value increasing from -10 to -4.61 between the manual tuning and the LQR cases. This reduces the stabilizing influence on the pendulum, resulting in larger deviations and therefore a higher error in the pendulum.

Effects of Different Q Values for LQR

The diagonal values of the Q matrix inversely corresponded to the weight of the system's states' errors of θ_1, θ_2, $\dot{\theta}_1$, and $\dot{\theta}_2$. The larger the values in the corresponding state's index, the more weight that state's performance had, resulting in the system to skew in performance to achieve a better output accuracy for that certain state. Changing the first index Q value of 70 that resulted in a K_1 gain value of -12.4486 in Case 1, to a Q value of 30 in that resulted in a K_1 gain value of -8.1495 in Case 2 results in a 12% RMSTE platen rotation accuracy decreases due to less weight on the platen rotation position. However, for the RMTSE values for the pendulum rotation position, the RMTSE pendulum rotation accuracy increases by 75%, from the second diagonal Q value of 40 that resulted in a K_2 gain value of -4.6143 in Case 1, to a Q value of 200 in that resulted in a K_2 gain value of -17.2448 in Case 2. The maximum pendulum angle also decreases following the Q matrix value changes from Case 1 to Case 2, from 50.1° to 40.3°, a 19.5% decrease with an increase in rise time from 0.483 to 0.497 seconds, a 2.82% increase. If there also was a criterion to have rise time minimized, the first diagonal index of the Q matrix would be increased as the rise time is dependent on the platen rotation velocity or gear motor actuation effort.

Effects of Different R Values for LQR

Between the initial LQR Case 1 and 3, the R value decreased from 1 to 0.1. This shifted the RMS command from 230 bits to 1126 bits, a 390% increase. LQR Case 4 increased the R value to 10, which shifted the RMS command to 106.2 bits, a 53.8% decrease from the initial case. This demonstrates how increasing the R value for the platen rotation decreases the RMS command and vice versa. Therefore, if the controller is too strong, the R value should be increased to reduce control input usage. This is supported by an increase in rise time of 2.90% between LQR Cases 1 and 3, going from 0.483 s to 0.497 s, as R was increased from 0.1 to 1. However, while the maximum pendulum angle was the lowest at an R value of 10, 46.7% lower than the angle at an R value of 1, shifting from 26.7° to 50.1°, it also had a higher RMSTE pendulum, being 47.3°, or an 42.9% increase from the initial LQR case 1 of 33.1°. Increases in R should be kept limited, as significant changes lead to a less accurate platen rotation.

Improvements

The pendulum breakaway socket did not provide adequate security to the pendulum encoder. During experimental runs, the pendulum would frequently detach itself from the assembly and result in an unscheduled maintenance session as the socket itself would break and require a recalibration of the system. An improvement to this interface would to add a set screw, a breakaway socket with a lip on the back to clip onto the encoder, or if no modifications can be made for the encoder interface itself, increase the amount of material in the center of the breakaway socket which would increase its resilience against impacts due to abrupt accelerations by the platen system detaching the pendulum.

V. References

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